10. Prove that $\sin\frac{\theta}{2}\sin\frac{7\theta}{2}+\sin\frac{3\theta}{2}\sin\frac{11\theta}{2}=\sin 2\theta\sin 5\theta$.
11. Prove that $\cos(30^\circ-A)\cos(30^\circ+A)+\cos(45^\circ-A)\cos(45^\circ+A)=\cos 2A+\frac{1}{4}$.
12. Prove that $\dfrac{\sin x+\sin 3x+\sin 5x+\sin 7x}{\cos x+\cos 3x+\cos 5x+\cos 7x}=\tan 4x$.
13. Prove that $\dfrac{\sin(4A-2B)+\sin(4B-2A)}{\cos(4A-2B)+\cos(4B-2A)}=\tan(A+B)$.
14. Show that $\cot(A+15^\circ)-\tan(A-15^\circ)=\dfrac{4\cos 2A}{1+2\sin 2A}$.

3.5.4 Conditional Trigonometric Identities

We know that trigonometric identities are true for all admissible values of the angle involved. There are some trigonometric identities which satisfy the given additional conditions. Such identities are called conditional trigonometric identities.

In this section, we shall make use of the relations obtained in the earlier sections to establish some conditional identities based on some relationship.

Example 3.39 If $A+B+C=\pi$, prove the following

(i) $\cos A+\cos B+\cos C=1+4\sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)$

(ii) $\sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)\le\frac{1}{8}$

(iii) $1<\cos A+\cos B+\cos C\le\frac{3}{2}$

Solution:

(i) $$\begin{aligned}\cos A+\cos B+\cos C &= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)+\cos C\\ &= 2\cos\left(\frac{\pi}{2}-\frac{C}{2}\right)\cos\left(\frac{A}{2}-\frac{B}{2}\right)+\cos C \quad \left(\frac{A+B+C}{2}=\frac{\pi}{2}\right)\\ &= 2\sin\left(\frac{C}{2}\right)\cos\left(\frac{A}{2}-\frac{B}{2}\right)+1-2\sin^2\left(\frac{C}{2}\right)\\ &= 1+2\sin\left(\frac{C}{2}\right)\left[\cos\left(\frac{A}{2}-\frac{B}{2}\right)-\sin\left(\frac{C}{2}\right)\right]\\ &= 1+2\sin\left(\frac{C}{2}\right)\left[\cos\left(\frac{A}{2}-\frac{B}{2}\right)-\cos\left(\frac{\pi}{2}-\frac{C}{2}\right)\right]\\ &= 1+2\sin\left(\frac{C}{2}\right)\left[\cos\left(\frac{A}{2}-\frac{B}{2}\right)-\cos\left(\frac{A}{2}+\frac{B}{2}\right)\right]\\ &= 1+4\sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)\end{aligned}$$

(ii) Let $$\begin{aligned}u &= \sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)\\ &= -\frac{1}{2}\left[\cos\left(\frac{A+B}{2}\right)-\cos\left(\frac{A-B}{2}\right)\right]\sin\frac{C}{2}\\ &= -\frac{1}{2}\left[\cos\left(\frac{A+B}{2}\right)-\cos\left(\frac{A-B}{2}\right)\right]\cos\frac{A+B}{2}\end{aligned}$$